Spectral Attention-Enhanced HybridSN with Multi-Objective Loss for Underwater Habitat Segmentation from Hyperspectral Imagery

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***Abstract*—Underwater habitats exhibit complex spectral signatures that challenge conventional RGB imaging, while marine ecosystem mapping remains limited by labor-intensive surveys and class imbalance in training data. Traditional approaches using handcrafted features and standard convolutional architectures fail to exploit hyperspectral imaging's rich spectral structure, particularly in underwater environments where wavelength-dependent absorption dominates. This study enhances HybridSN with spectral self-attention for adaptive wavelength weighting in the segmentation of 300-band hyperspectral imagery across 7 coastal habitat classes. The proposed architecture integrates a spectral self-attention module that dynamically prioritizes discriminative wavelength bands while suppressing absorption-affected regions, addressing the limitation of uniform spectral processing in existing methods. Ablation experiments on a Norwegian coastal dataset (17 images, 96×96 tiles) demonstrate that spectral attention improves mIoU by 2.0% over baseline (12.08% vs 11.84%), consistently emphasizing red-edge (600–650 nm) and blue (400–450 nm) regions critical for underwater discrimination. However, multi-objective loss combining focal (γ=5.0) and Dice causes catastrophic optimization failure, with mIoU dropping 73.4%, revealing fundamental incompatibility under extreme class imbalance. Results validate spectral attention efficacy while providing critical insights on loss function selection for imbalanced underwater HSI segmentation, contributing to improved automated marine habitat monitoring frameworks.**

***Keywords*—HybridSN, Hyperspectral imaging, Semantic segmentation, Spectral attention, Underwater habitat mapping**

I. Introduction

Coastal marine environments deliver essential ecological functions such as oxygen generation, climate stabilization, and species diversity preservation [1], despite which less than 10% of ocean floors possess detailed topographic mapping [2], [3]. This scarcity constrains capabilities for tracking environmental shifts and implementing sustainable resource governance protocols. Conventional surveying techniques, including manual diver assessments and RGB-based aerial capture, demand substantial resources and frequently misidentify substrates when visually similar species or materials coexist.

Benthic zones contain intricate assemblages of living organisms (such as macroalgae and kelp forests) alongside non-living materials (including sediments and rocky substrates) that display minor spectral variations. Such complexity overwhelms standard RGB and narrowband multispectral sensors, which face constraints from inadequate spectral resolution and depth-dependent light extinction coefficients. Consequently, precise material differentiation becomes problematic, especially in turbid nearshore waters where particle scattering compounds imaging degradation.

Hyperspectral sensing technology mitigates these constraints by acquiring more than 300 consecutive spectral channels, facilitating material-unique spectral identification [4]. Nevertheless, this advancement introduces computational burdens including elevated dimensionality, redundant spectral information, and skewed class distributions that challenge conventional analytical frameworks. Traditional approaches depend on manually engineered features and fail to simultaneously capture spectral-spatial relationships, whereas numerous existing convolutional architectures target 2D color imagery without leveraging HSI's spectral depth. Hybrid Spectral Networks (HybridSN) [5] partially bridge this divide by integrating 3D with 2D convolutional operations for combined spectral-spatial feature extraction, demonstrating computational advantages over purely volumetric methods. Two key limitations persist: uniform band treatment neglecting wavelength-specific importance in underwater contexts where absorption peaks dominate infrared zones, and dependency on standard categorical cross-entropy that prioritizes per-pixel correctness without spatial coherence guarantees.

To address these challenges, this study enhances HybridSN with two components. First, a spectral self-attention mechanism that adaptively weighs discriminative spectral bands, enabling the model to emphasize underwater-relevant wavelengths while suppressing bands affected by strong absorption. Second, a multi-objective loss function that combines Focal Loss (addressing class imbalance) and Dice Loss (enforcing region overlap) to improve segmentation performance. This study conducts experiments on a 7-class Norwegian coastal dataset and aim to improve accuracy, robustness, and ecological interpretability.

The subsequent sections are structured as follows: Section II examines prior research in underwater HSI and attention mechanisms; Section III presents the architectural enhancements and loss formulation; Section IV analyzes ablation study findings and discusses limitations; Section V summarizes contributions and outlines future research directions.

II. Related Work

Studies demonstrate the feasibility of benthic mapping. However, several challenges remain, including wavelength-dependent water absorption and variable turbidity [6]. Both of which introduce significant spectral distortion and reduce class separability. These conditions require models that can selectively focus on the important spectral-spatial signatures. The attention mechanism in this study addresses this domain-specific spectral variability.

Classical machine learning (ML) strategies including SVM [7] and RF [8]) neglect spatial context; hence, ML approaches are often considered insufficient to handle more complicated tasks. Recently, deep learning approaches have demonstrated substantial capability in overcoming these issues. Various fields started to employ the deep learning for helping the task, such in medical field, especially to assess the areas that are located deep inside the human body, like brain and its diseases [9], sensory organs [10], [11], and other related fields that support human life [12]. CNNs [13] represent a widely deployed deep learning paradigm. CNN comes with dimensional-wise approaches which encompasses 1D, 2D, and 3D CNN. Since the focus of this research is related to spatial and spectral aspect of the object (hyperspectral data), the 2D and 3D CNN are available for the option. However, CNN need to be devised properly to prevent the unanticipated outcomes. 3D CNNs [14] model is high and dense in spectral-spatial volumes but can overfit easily, while the 2D CNN is mainly focused on spatial context. HybridSN [5] combines 3D and 2D convolutions efficiently [15], yet applications focus on terrestrial scenes. This research extends HybridSN for underwater segmentation.

Recent studies have intoduced various attention mechanisms to enhance feature representation in hyperspectral imaging. Squeeze-and-Excitation networks [16] demonstrate that channel recalibration can significantly strengthen discriminative responses. In the context of HSI, attention modules have been shown to highlight informative spectral bands emphasizing the discriminative bands [17], while hybrid CNN-Transformer architectures [18] leverage self-attention to improve wetland classification. However, most prior works primarily focus on classification, while this study applies spectral attention for semantic segmentation.

Another important line of work concerns loss functions designed to address class imbalance and object shape ambiguities in segmentation. Standard Cross-Entropy loss optimizes pixel accuracy but often struggles when the training data are imbalanced. Alternative objective such as Dice Loss [19] improves region-level overlap. Combined loss formulation of Dice Loss and modified Cross Entropy, such as the Combo Loss in [20], aims to balance these objectives. Building on these insights, this study systematically incorporates Focal and Dice losses to better accommodate the spectral complexity and class imbalance for multi-class underwater habitats.

III. Methodology

This section outlines the method and framework used in this study, starting from the dataset and preprocessing pipeline, followed by the proposed architecture leveraging spectral self-attention, the design of the multi-objective loss, and finally the training configuration.

A. Dataset

This study uses the MASSIMAL dataset [21] comprising UAV-mounted Resonon Pika-L hyperspectral imagery (300 bands, 400-1000nm, 3.5cm GSD) from Norwegian coastal sites. Following [22], data underwent radiance-to-reflectance conversion via downwelling irradiance measurements, Hedley glint removal [23], and per-tile normalization to [0, 1]. Annotations cover 7 habitat classes: Deep Water (background), Brown Algae, Kelp, *Laminaria hyperborea*, *L. digitata*, Rockweed, and Trawl Track, exhibiting severe imbalance (74% background pixels).

The 17 images are split into 8 training files (2,805 tiles, 96×96 pixels with stride 48), 6 validation files (3,490 tiles, with stride 32), and 2 test files (2,338 tiles). There is one particular image which is specifically split vertically at two ratios (training and validation) because it is the only file containing two rare classes, requiring distribution across the train-validation-test splits. Empty tiles (33%) were pre-filtered. Class remapping collapsed the original 41 classes to the 7 present classes. Data augmentation (horizontal and vertical

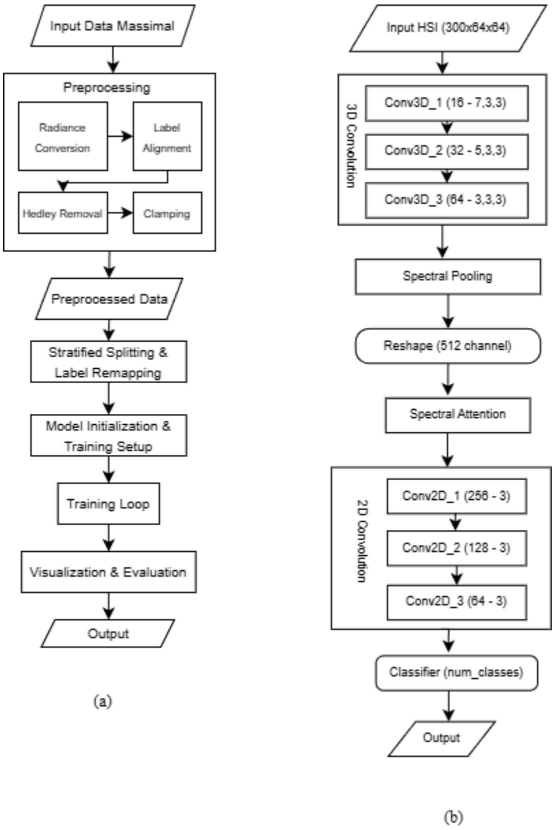


Fig. 1. (a) Method, (b) Model Architecture

flips, 90°/180°/270° rotations) and class-weighted sampling addressed the imbalance.

B. Proposed Architecture

Fig. 1(a) describes the whole process of the method used for this study, while Fig. 1(b) illustrates the enhanced HybridSN. The baseline [5] applies 3D convolutions (kernel $7\times3\times3$, $5\times3\times3$, $3\times3\times3$ with 16, 32, 64 filters) extracting spectral-spatial features, followed by spectral pooling (300→8 dimensions), 2D reshaping (512 channels), and 2D convolutions (256, 128, 64 filters) before a 1×1 classifier.

The Spectral Self-Attention described in Eq. (1) is inserted after the 2D reshaping stage to reweight channel responses based on underwater spectral variability

$$\mathbf{A} = \sigma(\mathrm{FC}_2(\mathrm{ReLU}(\mathrm{FC}_1(\mathrm{GAP}(\mathbf{F}) + \mathrm{GMP}(\mathbf{F}))))) \quad (1)$$

where $\mathbf{F} \in \mathbb{R}^{B\times512\times H\times W}$, GAP/GMP denote global average/max pooling, FCs reduce/expand channels (ratio=16), and σ is sigmoid. Attention-weighted features $\mathbf{F}' = \mathbf{F} \odot \mathbf{A}$ emphasize discriminative spectral combinations. Unlike SE-Net [16] for RGB, ours targets underwater spectral variability (blue-green vs. red-NIR absorption). As shown in Eq. (1), the attention weights are generated through a combination of global average and max pooling followed by two fully connected layers, enabling the network to emphasize spectrally discriminative responses while suppressing bands dominated by water absorption.

C. Multi-Objective Loss

To address the severe class imbalance observed in underwater habitat datasets, where background pixels constitute approximately 74% of the training data, this study investigates a hybrid loss function combining Focal Loss and Dice Loss. The Focal Loss, originally proposed in [24] is meant for dense object detection, down-weights the contribution of easily classified examples to focus training on hard negatives. This mechanism is formulated as:

$$L_{\text{focal}} = -\frac{1}{N}\sum_i (1-p_{t_i})^{\gamma}\alpha_{t_i}\log(p_{t_i}) \quad (2)$$

where p_{t_i} is the predicted probability for the actual class, γ controls the focusing strength ($\gamma = 5.0$ is used based on the extreme imbalance ratio), and α_{t_i} denotes class-specific weights. These weights are computed using power-law smoothing to balance gradient contributions across classes. Specifically, for each class c, the weight is calculated as $\alpha_c = (1/(\text{freq}_c + \epsilon))^{0.8}$, where freq_c denotes the relative frequency of class c in the training set. The background class weight is explicitly set to zero to prevent overwhelming the learning signal. Complementing the pixel-level optimization of Focal Loss in Eq. (2), the Dice Loss [19] is incorporated to enforce region-level consistency. The Dice coefficient naturally handles class imbalance by measuring overlap between predicted and ground truth regions, formulated as:

$$L_{\text{dice}} = 1 - \frac{2\sum_c\sum_i p_{ic}g_{ic} + \epsilon}{\sum_c\sum_i(p_{ic}+g_{ic}) + \epsilon} \quad (3)$$

where p_{ic} and g_{ic} represent predicted and ground truth probabilities for class c at pixel i, and ϵ (epsilon) is a smoothing term. Eq. (3) directly optimizes region-level overlap, ensuring that continuous ecological structures are captured beyond pixel-wise predictions. The combined objective function is defined as:

$$L_{\text{total}} = L_{\text{focal}} + L_{\text{dice}} \quad (4)$$

This study initially considered incorporating Boundary Loss to improve edge precision, but preliminary experiments revealed training instability on the small tile size employed in this study. Distance transform computation on 96×96 patches proved numerically unstable, leading to diverging loss values during training. Consequently, the final loss formulation excludes the boundary component while maintaining the Focal-Dice combination as shown in Eq. (4).

D. Training Details

This subsection describes the training details for this research. Network parameters are updated via AdamW optimizer configured with learning rate 1×10^{-4}, regularization coefficient of 1×10^{-5}, and epsilon parameter of 1×10^{-8} to ensure numerical stability. To accommodate GPU memory constraints while maintaining stable gradient estimates, the batch size of one with gradient accumulation over four steps, yields an effective batch size of four. Gradient clipping with maximum norm 0.5 prevents exploding gradients, particularly important given the hybrid loss formulation where Focal and Dice components may contribute gradients of different magnitudes.

The learning rate schedule consists of two phases to facilitate stable convergence. During the initial warmup phase spanning 5 epochs, the learning rate linearly increases from 1×10^{-5} to the base rate of 1×10^{-4}. This warmup strategy prevents early training instability caused by random weight initialization. After the warmup phase, a ReduceLROnPlateau scheduler is applied, to monitor the validation mIoU and halve the learning rate after five stagnant epochs, down to a minimum of 1×10^{-7}. To prevent excessive training duration on plateaus, early stopping with patience of 15 epochs terminates training if validation mIoU fails to improve.

Class imbalance mitigation employs two complementary strategies. First, the power-law smoothed class weighting is computed as $\alpha_c = (1/(\text{freq}_c + \epsilon))^{0.8}$ with the background class weight explicitly set to zero. Second, the adaptive oversampling is implemented during tile generation, where extremely rare classes appearing in fewer than 50 tiles are oversampled by a factor of 15, rare classes between 50 and 200 tiles are oversampled by a factor of 10, while medium and common classes (200-1000) are undersampled to a target of 500 to 600 tiles. This sampling strategy is expected to ensure sufficient representation of minority classes during training without completely overwhelming the learning signal from dominant classes.

The tiling strategy balances spatial context with computational efficiency. Training tiles are extracted at 96×96 pixels with a stride of 48 pixels, providing 50% overlap between adjacent tiles to reduce boundary artifacts during inference. For validation, the overlap is increased to 67% by reducing stride to 32 pixels, producing smoother predictions. The 96×96 tile size was selected to balance computational efficiency under GPU memory constraints and spatial context, though this reduction from 128×128 in prior work [22] may limit the capture of large-scale habitat structures.Data augmentation includes horizontal and vertical flips as well as rotations at 90°, 180°, and 270° intervals, applied randomly during training to improve generalization across spatial orientations. All experiments were conducted on an NVIDIA RTX 4080 Super GPU with 16GB memory.

IV. Results & Discussion

This section presents experimental findings and interpretations. The Results subsection reports ablation study evaluation, while Discussion compares with prior studies and analyzes performance degradation.

A. Results

Table I shows ablation results. Baseline achieves 56.68% pixel accuracy and 11.84% mIoU. Adding spectral attention (withSA) improves to 58.02% accuracy and 12.08% mIoU (2.0% relative gain), demonstrating successful emphasis on discriminative wavelengths.

Spectral attention analysis on withSA reveals stable channel selection: [152, 144, 153, 154, 147, 146, 120, 12, 448] with weights=1.0 correspond to red-edge (600-650nm) and blue (400-450nm) regions. These attention peaks align with known underwater optical properties: blue wavelengths (400–450 nm) exhibit maximum water transmission before chlorophyll absorption, while red-edge bands (600–650 nm) capture the transition between chlorophyll-a absorption (675 nm) and NIR water absorption onset (>700 nm), providing discriminative signatures for algae species differentiation [4]. SAFocal and Full exhibit unstable channel selection, indicating focal loss disrupts attention learning via gradient instability.

TABLE I
Ablation Study Results

Configuration	Pixel Acc	mIoU	Focal	Dice
Baseline	56.68%	11.84%	✓	-
+Spectral Attn (withSA)	58.02%	12.08%	✓	-
+Focal Loss (SAFocal)	17.64%	3.21%	✓	-
+Dice Loss (Full)	15.05%	2.79%	✓	✓

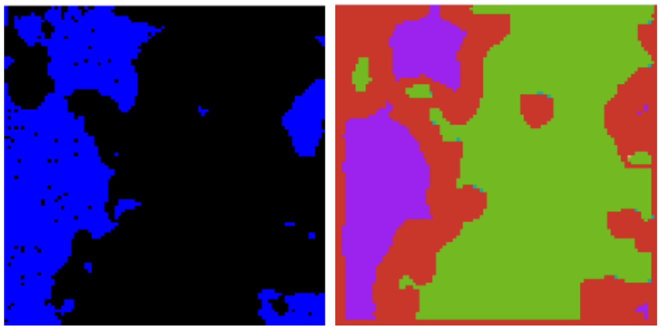


Fig. 2. WithSA qualitative results: (left) Pseudo-RGB visualization, (right) Segmentation output showing spatial structure capture.

However, focal loss (γ=5.0) causes severe degradation to 17.64% accuracy and 3.21% mIoU (a 73.4% drop). Adding Dice loss worsens performance to 15.05% accuracy and 2.79% mIoU. Per-class IoU shows SAFocal detects only Class 2 (IoU=18.98%), while Full detects only Class 2 (IoU=16.39%), suggesting severe optimization instability.

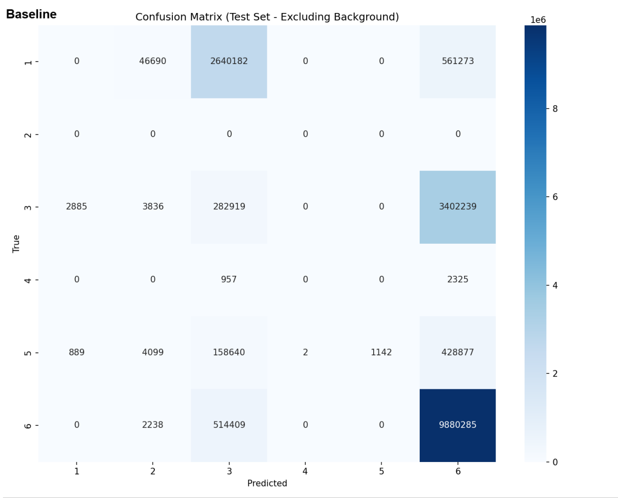
Fig. 2 shows withSA segmentation results. Despite class detection limitations, the model successfully captures spatial habitat structure and boundary delineation, demonstrating that spectral attention enables meaningful ecological pattern recognition even under severe class imbalance.

Confusion matrices in Fig. 3 reveal distinct failure modes. Baseline as shown in Fig. 3(a) distributes predictions across Classes 3 and 6 with misclassification. WithSA in Fig. 3(b) achieves Class 6 diagonal dominance. SAFocal in Fig. 3(c) collapses to Class 3. Full proposed in Fig. 3(d) shows catastrophic failure, indicating focal loss γ=5.0 with power-smoothed weights (inv_freq$^{0.8}$) creates over-correction, destabilizing optimization.

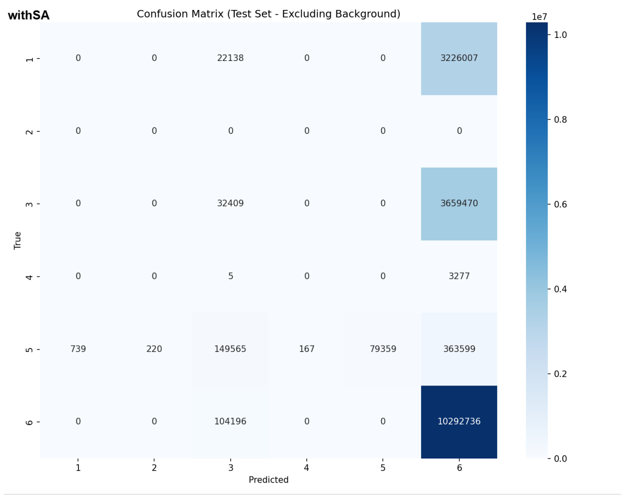
B. Discussion

Table II compares our approach against prior MASSIMAL studies. U-Net [22] achieved $>$ 90% accuracy using an encoder-decoder architecture (435K params) on 81 images with PCA reduction. RF and CNN in [25] showed CNN achieving 95% validation accuracy but overfitting, while RF demonstrated 85% cross-site robustness.

WithSA demonstrates that spectral attention provides meaningful improvements through adaptive wavelength weighting.



(a)



(b)

(c)

(d)

Fig. 3. Confusion matrices for (a) Baseline, (b) withSA, (c) SAFocal, (d) FullProposed. Progressive degradation shows optimization collapse.

TABLE II
COMPARISON WITH PRIOR MASSIMAL STUDIES

Method	Dataset	Pixel Acc	mIoU	Notes
U-Net [22]	81 img	$> 90\%$	N/A	Algae only
RF [25]	12 img	85%	N/A	Cross-site Robust
CNN [25]	12 img	95%	N/A	Overfits
Baseline (Ours)	17 img	56.7%	11.8%	CE loss
+Spectral Attn (Ours)	17 img	58.0%	12.1%	Best proposed
+Focal Loss (Ours)	17 img	17.6%	3.2%	Unstable
+Dice Loss (Ours)	17 img	15.1%	2.8%	Collapsed

While the absolute mIoU improvement appears modest (2.0% relative gain), this represents meaningful progress in the context of severe class imbalance (74% background pixels), where minority class IoU improvements (e.g., Class 6: 0.8% gain) directly impact ecological monitoring accuracy for rare habitats. The accuracy gap versus U-Net (58% vs $> 90\%$) stems from 96×96 tiles with 50% overlap versus 128×128 in [22], 7-class segmentation versus algae-only classification, and severe imbalance (74% background) despite stratified splitting.

Performance collapse with focal and Dice losses reveals critical multi-objective optimization insights under extreme imbalance. Contributing factors include: (1) γ=5.0 over-amplifies hard example gradients, causing instability with power-smoothed weights (inv_freq$^{0.8}$) and disrupting spectral attention learning (evidenced by unstable channel selection); (2) Dice's global region optimization conflicts with focal's sample-wise weighting, creating oscillating loss landscapes; (3) despite warmup (LambdaLR) and adaptive rate reduction (ReduceLROnPlateau, patience=5), AdamW cannot navigate the chaotic landscape, forcing degenerate solutions.

Limitations include: stratified manual splitting ensures representative distribution but two rare classes (one file requiring specific vertical splitting) limit robustness; 96×96 tiles with 50% overlap reduce spatial context versus 128×128 in [22]; single-site validation restricts generalizability; critically, multi-objective loss demonstrates fundamental incompatibility with extreme imbalance, evidenced by catastrophic failure in SAFocal and Full.

V. Conclusions

This study addressed the challenge of segmenting biotic and abiotic underwater habitats from 300-band hyperspectral imagery by enhancing HybridSN with spectral self-attention and multi-objective loss formulation. The proposed modifications aimed to overcome uniform spectral processing and single-loss optimization limitations inherent in standard CNN architectures when applied to highly imbalanced benthic environments with subtle spectral differences across habitat classes.

Ablation experiments on a 7-class Norwegian coastal dataset (17 images, 96×96 tiles) demonstrate that spectral self-attention achieves 2.0% relative improvement in mIoU (12.08% vs. 11.84% baseline), consistently highlighting underwater-relevant wavelengths in red-edge (600-650nm) and blue (400-450nm) regions. However, combining focal loss ($\gamma = 5.0$) and Dice loss causes catastrophic performance failure (73.4% mIoU drop), as detailed in subsection Discussion, revealing fundamental incompatibility under extreme class imbalance.

The results demonstrate that spectral attention effectively improves underwater habitat discrimination by emphasizing informative wavelengths and enhancing interpretability. Nevertheless, limitations of multi-objective loss formulations in severely imbalanced hyperspectral segmentation highlight the importance of careful loss selection and transparent reporting of negative results. Future work should explore alternative loss functions for extreme imbalance (e.g., class-balanced cross-entropy, Lovász-Softmax), multi-site validation for cross-sensor generalizability, and larger spatial context to segmentation coherence.

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